

# PFC AI Design Agent Report Documentation Analysis & Improvement Master Plan

Based on study of four reference documents combined with the  
previously agreed Documentation Improvement Plan (June 2026)

*Document 1: EDGE 0059392A2 Full Magnetic Design Report*

*Document 2: FAN9672 Control Loop Design Calculations Rev 2.1*

*Document 3: Output Capacitor Calculation (Step 15 reference)*

*Document 4: PFC Design Report Steps 1–15 (current tool output)*

*Version 1.0 · June 2026 · PFC Engineering Team*

This document synthesises the key formatting, content, and pedagogical techniques observed in four reference engineering reports into a concrete implementation plan for the PFC AI Design Agent report generator. It supersedes and extends the Documentation Improvement Plan of June 2026.

## 1. What the Four Reference Documents Reveal

All four documents were written by engineers who designed the same PFC power stage using different tools and audiences in mind. Together they provide an exceptionally clear picture of what a best-in-class PFC design report looks like.

### 1.1 Document 1 — EDGE 0059392A2 Magnetic Design Report

**Character:** Step-by-step magnetic design with 24 numbered steps. Most technically detailed of the four. Written by a domain expert for peer review.

#### Key techniques to adopt:

- **Corrections table as Step 2:** Before any calculation, a table lists every error found in the previous design iteration and what was corrected. This is extraordinarily powerful for building trust — the reader immediately sees the report has been audited.
- **Material model validation section (Step 3):** Every Steinmetz constant and permeability model is validated against a specific datasheet reference point, with a ✓ mark. Example: " $a_{\text{max}} \cdot 0.1^2 \cdot 106 \cdot 100^{1.444} = 525.0 \text{ mW/cm}^3$  ✓ (reproduces datasheet anchor)." No reader can question whether the model is calibrated.
- **Tolerance columns in every table:** Loss and inductance tables show three columns: min / nominal / max based on  $AL \pm 8\%$  tolerance. This alone distinguishes a production-ready design report from a student exercise.
- **CCM/DCM boundary verification (Step 15):** Explicitly checks that CCM holds throughout the half cycle — validating a core assumption of the entire model.
- **Saturation margin step (Step 16):** "50% inductance not reached until  $2.8\times$  the operating peak current." A single sentence that tells the approval engineer everything.
- **Interleaving ripple cancellation as a dedicated step (Step 17):**  $K(D)$  explained and plotted. The report explicitly states "per-inductor flux and loss are NOT reduced by interleaving" — a common misconception addressed head-on.
- **Sensitivity/tolerance table (Step 21):** Lists  $AL$ , temperature, material variations and their effect on each output quantity. Quantifies what happens if a parameter is wrong.
- **Material validity table (Step 22):** Confirms the model assumptions hold at the operating point ( $\mu$  flat to 3 MHz at 70 kHz → ✓).
- **Validation and measurement plan (Step 23):** A table of bench tests:  $L$  vs  $I$  with DC bias fixture, thermal imaging, DCR measurement. The report tells you how to verify it.
- **The "double-hump" explanation (Step 11.2):** "At high line the peak occurs off-crest where  $V_{\text{in}} \approx 195 \text{ V}$  ( $D \approx 0.5$ ) — the double-hump signature." One sentence with a figure reference. Junior engineers will understand something that confuses many experienced designers.

### 1.2 Document 2 — FAN9672 Control Loop Design Calculations Rev 2.1

**Character:** Tutorial-quality control loop design. The most comprehensive and best-written of the four. Explicitly described as "both a verifiable engineering deliverable and a self-contained tutorial: every result is derived from first principles or a cited application note."

### Key techniques to adopt:

- **Executive Summary scorecard at the very top:** Before any calculation, a table shows every key result — crossover frequency, phase margin, gain margin, 120 Hz rejection — with the floor/target and a PASS/FAIL verdict. The approval engineer reads the first page and knows the status of the entire design.
- **Nomenclature table + Abbreviations table:** Every symbol defined with value and units. Abbreviations listed. Placed before the first calculation so the reader is never stopped by an undefined term.
- **CONCEPT box at the start of every step:** 2–4 sentences explaining WHY the calculation exists before showing any equation. This is the single most important structural element in the document.
- **PITFALL boxes:** Shaded boxes identifying the most common errors at each step. "PITFALL: Forgetting the  $4\times$  HV multiplier is the most common error at this step." These prevent mistakes in downstream designs.
- **DECISION bars:** Each calculation ends with a shaded Decision box recording what was chosen and why. "Decision:  $R_{CS} = 15.0\text{ m}\Omega$  — inside the overlap of both methods at  $K_{max} = 1.49$ ." The design rationale is captured permanently.
- **INSIGHT boxes:** Non-obvious engineering observations. "INSIGHT: Five parts is driven by ripple-current sharing, not capacitance alone." Makes the report educational, not just computational.
- **Three-path verification for G\_MOD (Step 7):** Three independent calculation methods that must all agree. If they disagree, the component values are inconsistent. A built-in self-audit that builds enormous confidence.
- **All 8 operating points verified in every table:** No cherry-picking of favorable conditions. The design is verified at 90, 110, 120, 132, 180, 210, 220, 264 Vac.
- **Document control block:** Title, document number, revision, status, author, classification. Makes the report a formal deliverable, not just a calculation sheet.
- **Revision history table:** What changed in each revision. Builds a traceable audit trail.
- **"How to Read This Report" section:** Brief guide to the document structure. Reduces cognitive load for the reader.
- **"How the steps fit together" roadmap:** Introduction section explaining the dependency chain between steps. Context before detail.
- **Pin-by-pin table with external network values:** Every IC pin listed with its function and the actual component value connected to it. A junior engineer can use this to build the circuit.
- **Reference citations by document number:** AN4165-D Eq. 39, AND9925-D Table 1. Not just "per datasheet" — exact equation numbers.
- **R\_GC / R\_LS sizing per AN4165-D (Step 8):** Components that previous revisions left as "application-set" were formally sized in Rev 2.1. This is the right pattern — no component is left unsized.

## 1.3 Document 3 — Output Capacitor Calculation

**Character:** Standalone capacitor sizing document. Uses the same CONCEPT / THEORY / DECISION / PITFALL / INSIGHT structure as Doc 2. Exemplary treatment of the lifetime model.

### Key techniques to adopt:

- **Introduction explaining WHY electrolytics wear out:** Physical basis before any formula. "An aluminium electrolytic ages because its liquid electrolyte slowly escapes through the rubber end-seal. The escape rate is a chemical reaction, so it follows the Arrhenius law: it roughly doubles for every 10 °C." This is perfect Layer 1 content.
- **THEORY boxes for model derivation:** Separate from the CONCEPT box. CONCEPT = why we need this; THEORY = the mathematical basis.
- **Numbered sub-steps within each step:** "Step 1: Hold-up capacitance: [equation]. Step 2: Denominator: [numerical substitution]." Every step is broken into atomic operations. No reader loses the thread.
- **PITFALL for frequency conversion (Step 15.12):** "Do not apply the LF ESR to the HF current. The HF component is divided by its larger frequency coefficient... skipping this over-states the heating by ~25%."
- **Three-method comparison with confidence band interpretation:** "Read the spread as a confidence band, not a contradiction. Method 2 is the conservative floor... Method 1 is the realistic expectation... Method 3 is the formal qualification method." This framing is essential for a decision-maker.
- **INSIGHT on why 5 caps beat 1 large cap (Step 15.4):** "Five parts is driven by ripple-current sharing, not capacitance alone: spreading 12.97 A across five cans keeps each one at well under half its rating, which is what gives the bank its very long life." This is the kind of insight that makes engineers better.
- **Voltage derating  $f(V)$  — Method 3 vs simplified:** "The size-based  $f(V) = 2.58\times$  is well above the  $(450/390)^3 = 1.54\times$  used by Methods 1–2 — a 67% difference that is the main reason Method 3 reads highest." Explicitly quantifying the source of differences between methods prevents confusion.

## 1.4 Document 4 — PFC Design Report Steps 1–15 (Current Tool Output)

**Character:** This is the current tool's output report — the baseline we are improving. Comparing it to Documents 1–3 reveals the specific gaps.

### What it does well:

- Steps 1–12 cover the full PFC power stage with correct calculations
- Steps 13–14 have good table structure with min/nom/max
- CONCEPT boxes are present in some steps
- Figures referenced with captions
- Step 15 (capacitor) has good basic structure

### Specific gaps versus the reference documents:

- **No Executive Summary scorecard** — compared to Doc 2's first-page all-results table
- **No Nomenclature table** — symbols appear without prior definition in many steps
- **No PITFALL boxes** — common errors not flagged anywhere
- **No DECISION boxes** — design rationale not captured
- **No Corrections / Methodology table** — unlike Doc 1 which opens with what changed
- **No Material Model Validation section** — Steinmetz constants used without anchor proof
- **No Tolerance columns** (min/nom/max) in loss tables
- **No CCM/DCM boundary verification**

- **No Saturation and Inductance Margin step**
- **No Interleaving Ripple Cancellation step** (K(D) derivation missing)
- **No Sensitivity/Tolerance table**
- **No Bench Measurement/Validation Plan**
- **No "How to Read This Report" section**
- **No Revision History**
- **No Document Control block**
- **Step 16 (Control Loop) is entirely missing Doc 2's depth** — no CONCEPT primers, no PITFALL boxes, no three-path verification, no pin table

## 2. The Five Structural Elements to Implement

All four reference documents converge on five structural elements that appear consistently across every technically strong step. These are not cosmetic — they fundamentally determine whether a reader can follow, trust, and use the report.

### 2.1 The CONCEPT Box

Appears at the start of every step. 2–4 sentences. Answers: *Why does this calculation exist? What would go wrong if we skipped it?* Requires zero prior knowledge of the specific calculation.

#### CONCEPT Current Loop Crossover Frequency

The current loop must be fast enough to force the inductor current to track the rectified-sine reference on a sub-line-cycle timescale, yet slow enough to avoid noise amplification at the switching frequency. Setting  $f_{ci} = f_{sw}/8$  is the canonical starting point: the loop has enough margin from the Nyquist frequency ( $f_{sw}/2$ ) to give robust phase margin, while still tracking the 120 Hz current envelope with negligible error.

### 2.2 The THEORY Box

Used when a governing equation requires physical derivation before it can be applied. Separate from CONCEPT. CONCEPT answers "why"; THEORY answers "what equation and where does it come from."

#### THEORY Hold-up Energy Balance

During a line dropout the capacitor must supply the full load while its voltage falls from  $V_{out}$  to  $V_{dc,min}$ . Setting the stored-energy change equal to the energy delivered to the load:  $\frac{1}{2}C \cdot (V_{out}^2 - V_{dc,min}^2) = P_{out} \cdot t_{hold}$ . Solving for C:  $C_{hold} = 2 \cdot P_{out} \cdot t_{hold} / (V_{out}^2 - V_{dc,min}^2)$ . Note that SEMI F47 sets  $V_{dc,min} = 0.70 \cdot V_{dc,nom}$  for the 200 ms voltage-sag condition.

### 2.3 The PITFALL Box

Documents the most common mistake at this specific step. Found by experience, not derivation. Invaluable for reproducibility and peer review.

#### PITFALL iGSE Correction — Wrong Waveform Assumption

Applying the Steinmetz equation at peak B using the sinusoidal-Steinmetz formula (without the  $F(D)$  correction) overestimates core loss at high line ( $D \approx 0.05$ , where  $F(D) \approx 1.63$ ) by up to 63%, and underestimates at low line. The iGSE  $F(D)$  correction is not optional for a trapezoidal waveform PFC inductor — this was the largest single error in previous versions of this calculation.

### 2.4 The DECISION Box

Records exactly what was chosen, why it was chosen, and the margin it provides. This is the audit trail the approval engineer needs.

**DECISION Selected Core: EDGE 0059392A2 × 2 Stack**

Selected over the 3-stack option because: (1) window fill factor 70.4% vs 85%+ for 3-stack (within practical winding limit), (2) installed height 42.2 mm vs 62 mm for 3-stack (fits 1U chassis), (3) total loss 5.88 W max vs 4.9 W for 3-stack (acceptable 20% penalty for the size benefit). The 2-stack design provides 25% temperature margin and 2.3× saturation margin at all operating points.

## 2.5 The INSIGHT Box

Non-obvious observations that make the report educational. These are the sentences that junior engineers remember years later.

**INSIGHT The Double-Hump Signature at High Line**

At high line (180–264 Vac), the core loss does not peak at the line crest (where current is maximum). Instead, it peaks at the two instants where  $V_{in} = V_{out}/2$  ( $D = 0.5$ ), giving two loss humps straddling the crest. This is because core loss scales with  $B_{ac,pk}$ , which is maximum when  $V_{in} \cdot D$  is maximum, and  $D(1-D)$  is maximum at  $D = 0.5$ . A designer who uses crest-point  $P_{core}$  for high-line thermal sizing is using the wrong data point.

### 3. Numbered Sub-Steps: The Calculation Format

Documents 2 and 3 use a specific format for every calculation that eliminates ambiguity: each step has a conceptual introduction, then numbered sub-steps showing the equation, the substitution, and the numerical result, then a Decision.

#### 3.1 The Template

Every calculation step follows this exact pattern:

| Element               | Content                                                       | Purpose                                                    |
|-----------------------|---------------------------------------------------------------|------------------------------------------------------------|
| CONCEPT               | Why this calculation exists (2–4 sentences, no equations)     | Orients the reader before they see any math                |
| THEORY (optional)     | Derivation of the governing equation from first principles    | Required when the equation is not self-evident             |
| Governing equation    | Symbolic form with every variable defined in the nomenclature | Allows independent reproduction                            |
| Step 1, Step 2, ...   | Numerical substitution, one operation per sub-step            | Reader can follow the arithmetic without losing the thread |
| PITFALL (conditional) | The most common mistake at this specific step                 | Prevents known errors in downstream designs                |
| DECISION              | Selected value, standard rounding, margin provided            | Captures the design rationale permanently                  |
| INSIGHT (conditional) | Non-obvious observation about the result                      | Makes the report educational                               |

#### 3.2 Example — Step 15.2: Hold-up Capacitance (from Doc 3)

The following shows how Document 3 applies this template to a specific calculation. This level of detail is what we need to achieve in every step of our report.

##### Step 15.2 — C<sub>out</sub> from Hold-up

###### THEORY Hold-up Energy Balance

During a line dropout the capacitor supplies the full load while its voltage falls from V<sub>out</sub> to V<sub>dc,min</sub>. Equating stored-energy change to energy delivered:

$$C_{out,hold} = (2 \cdot P_{out} \cdot t_{hold}) / (V_{out}^2 - V_{dc,min}^2)$$

**Step 1** Denominator:

$$V_{out}^2 - V_{dc,min}^2 = 390^2 - 290^2 = 152100 - 84100 = 68000 \text{ V}^2$$

**Step 2** Hold-up capacitance:

$$C_{out,hold} = 2 \times 3600 \times 0.020 / 68000 = 0.002118 \text{ F} = 2117.6 \text{ }\mu\text{F}$$

**DECISION** C<sub>out,hold</sub> = 2117.6  $\mu$ F

This is the larger of the two sizing demands (hold-up vs ripple voltage). The hold-up requirement governs the bank size. The 10 ms hold-up time is set by SEMI F47 Class 1 — 20 ms at 70% Vnom.

## 4. New Steps and Sections to Add to the Report

Comparing the four reference documents against our current Steps 1–16 reveals specific content blocks that are completely missing. Each one below is a concrete item to implement.

| New Addition                        | Source Doc           | Where to Insert         | Content Description                                                                                                          |
|-------------------------------------|----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Design Corrections Table            | Doc 1, Step 2        | Start of Step 13        | Table of all corrections applied from previous design iteration. Builds trust in the report.                                 |
| Material Model Validation           | Doc 1, Step 3        | Step 13.1               | Validate Steinmetz constants a,b,c against specific datasheet data points with ✓ marks.                                      |
| iGSE F(D) Derivation                | Doc 1, Step 3.3      | Step 13.2               | Show the full F(D) formula, explain why it differs from sinusoidal Steinmetz, and compute F(D) at each corner.               |
| Tolerance Columns (min/nom/max)     | Doc 1, Tables        | Steps 13–14 tables      | All inductance and loss tables should show AL-min, AL-nom, AL-max columns.                                                   |
| CCM/DCM Boundary Check              | Doc 1, Step 15       | Step 14 end             | Verify CCM holds throughout the full half cycle. Identify where DCM appears (near zero-crossings at partial load).           |
| Saturation & Inductance Margin      | Doc 1, Step 16       | Step 13 end             | Show B_max vs B_sat margin. Show L vs I curve with operating point and 50%-L point marked.                                   |
| Interleaving Ripple Cancellation    | Doc 1, Step 17       | After Step 4            | K(D) formula, K(D) vs D plot, explain why per-inductor loss is NOT reduced, and input benefits.                              |
| Sensitivity / Tolerance Table       | Doc 1, Step 21       | Step 14 end             | Show effect of AL ±8%, temperature variation, material lot variation on all key outputs.                                     |
| Material Validity Table             | Doc 1, Step 22       | Step 13.1               | Confirm model assumptions: $\mu$ flat vs frequency at $f_{sw}$ , Steinmetz valid at operating B <sub>ac</sub> .              |
| Bench Measurement / Validation Plan | Doc 1, Step 23       | End of Step 14          | Table: what to measure, how (instrument + fixture), what it confirms. LCR+DC bias, thermal camera, VNA.                      |
| Executive Summary Scorecard         | Doc 2, Front         | Page 1 of report        | All key results with pass/fail verdict in one table. Current crossover, PM, core loss, $\Delta T$ , fill, saturation margin. |
| Nomenclature Table                  | Doc 2, Front         | After Executive Summary | Every symbol, its meaning, units, and value for this design. No undefined terms in the body.                                 |
| PITFALL Boxes Throughout            | Docs 2 & 3           | All steps               | One PITFALL box per calculation step where a common mistake exists. Identify from experience.                                |
| DECISION Boxes Throughout           | Docs 2 & 3           | End of every step       | Record selected value, standard value, margin, and rationale for every design decision.                                      |
| Three-Path G_MOD Verification       | Doc 2, Step 7        | Step 16 or new Step 6   | Verify gain modulator scaling with 3 independent methods. A/B ratio = 1.000 is the self-audit.                               |
| Pin-by-Pin IC Table                 | Doc 2, Section S.1   | Step 16                 | Every FAN9672 pin listed with external component values. Junior engineer can build the circuit.                              |
| Capacitor Three-Method Lifetime     | Doc 3, Steps 15.9–16 | Step 15                 | Three independent Arrhenius methods with explicit confidence band interpretation.                                            |

|                        |                 |            |                                                                                               |
|------------------------|-----------------|------------|-----------------------------------------------------------------------------------------------|
| Document Control Block | Doc 2,<br>Cover | Cover page | Document number, revision, status, author, date, classification. Formal deliverable metadata. |
| Revision History Table | Doc 2,<br>Cover | Cover page | What changed in each auto-generated revision. Enables traceability over design iterations.    |

## 5. Writing Style Guide — Adapted from Reference Documents

The four reference documents use a consistent style that is formal, active, precise, and pedagogical. The following rules distil what makes them work.

### 5.1 Sentence-Level Rules

- **State the equation, then substitute numbers on the same line.** " $C = 2 \cdot P \cdot t / (V_{\text{max}}^2 - V_{\text{min}}^2) = 2 \times 3600 \times 0.020 / 68000 = 2117.6 \mu\text{F}$ " — never on separate paragraphs.
- **Every equation gets a plain-English companion sentence.** After the formula: "This says the capacitor must supply the entire load power for  $t_{\text{hold}}$  seconds while its voltage drops by the stated amount."
- **Quantify comparisons.** Not " $B_{\text{max}}$  is well below saturation" but " $B_{\text{max}} = 0.43 \text{ T}$  versus  $B_{\text{sat}} \approx 1 \text{ T}$  — a  $2.3\times$  margin."
- **Name the governing principle.** Not "the temperature rises" but "the SA power-law  $\Delta T = (P \times 1000 / \text{SA})^{0.833}$  gives  $\Delta T = 28.6^\circ\text{C}$ ."
- **Use consistent notation throughout.** If a symbol is defined in the Nomenclature table, use exactly that symbol everywhere — no synonyms, no abbreviations.
- **Passive voice only in THEORY boxes.** Active voice everywhere else: "We select  $R2 = 143 \text{ k}\Omega$ " not " $R2$  is selected."

### 5.2 Table Design Rules

- **Every table has a pass/fail or assessment column** when the result is a criteria check.
- **Highlight the governing (worst-case) row** in amber or bold.
- **Show min / nominal / max** for any value with tolerance.
- **Include units in the column header**, not in every cell.
- **Reference every table figure in the narrative.** "Step 1 yields the results in Table X.Y."

### 5.3 Figure Rules (from all four documents)

- **Every figure has a number and a descriptive caption** (not just a title): "Figure 11.1 —  $P_{\text{core}}(t)$  (iGSE, max-anchored) over half line cycle, all input voltages."
- **Reference every figure in the narrative** before or on the same page it appears.
- **For multi-curve plots, label the worst-case curve** explicitly.
- **Add a horizontal line at the limit/target** on every margin/safety plot.

## 6. Combined Implementation Priority — Master Checklist

This table merges the previously agreed Documentation Improvement Plan with the new items identified from the four reference documents. Items are ranked by impact on readability and trustworthiness of the report.

| Pri. | Item                                                                 | Report Sections  | Source   | Effort |
|------|----------------------------------------------------------------------|------------------|----------|--------|
| 1    | Executive Summary scorecard (all results, first page)                | Cover page       | Doc 2    | Low    |
| 1    | CONCEPT + DECISION boxes at every step                               | All steps        | Docs 2,3 | High   |
| 1    | Nomenclature table (all symbols defined)                             | Front matter     | Doc 2    | Low    |
| 2    | PITFALL boxes at high-risk steps                                     | Steps 7,13,15,16 | Docs 2,3 | Medium |
| 2    | Numbered sub-steps (Step 1, Step 2) in every calculation             | All steps        | Docs 2,3 | High   |
| 2    | Material model validation (Steinmetz anchor proof)                   | Step 13.1        | Doc 1    | Low    |
| 2    | Tolerance columns min/nom/max in inductance & loss tables            | Steps 13,14      | Doc 1    | Medium |
| 2    | iGSE F(D) formula derivation with F(D) table at design corners       | Step 13.2        | Docs 1,4 | Low    |
| 3    | CCM/DCM boundary verification step                                   | Step 14 end      | Doc 1    | Low    |
| 3    | Saturation & inductance margin step (L vs I curve)                   | Step 13 end      | Doc 1    | Low    |
| 3    | Interleaving ripple cancellation K(D) — dedicated section            | After Step 4     | Docs 1,4 | Medium |
| 3    | Three-method capacitor lifetime with confidence band interpretation  | Step 15.9–15.16  | Docs 3,4 | Medium |
| 3    | Step 16 — CONCEPT boxes + PITFALL + control architecture intro       | Step 16          | Doc 2    | High   |
| 3    | Step 16 — Pin-by-pin IC table with component values                  | Step 16          | Doc 2    | Medium |
| 4    | Sensitivity/tolerance table (AL, temperature, material variation)    | Step 14 end      | Doc 1    | Medium |
| 4    | Material validity check table (model assumptions at operating point) | Step 13.1        | Doc 1    | Low    |
| 4    | Bench measurement / validation plan                                  | End of Step 14   | Doc 1    | Low    |
| 4    | Document control block + revision history table                      | Cover page       | Doc 2    | Low    |
| 4    | Design corrections table (what changed from previous iteration)      | Front matter     | Doc 1    | Low    |
| 4    | Three-path G_MOD verification (self-audit of gain-modulator scaling) | Step 16          | Doc 2    | Medium |



## 7. Open Questions — Answers Needed Before Implementation

The following decisions must be confirmed before writing the new report content. Each has a clear implication for the writing style and depth of every section.

| # | Question                                                                                                                                                                                           | Options                                                                                                                | Implication                                                                                                                                                               |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Reference citation style?</b><br>The reference documents cite specific app-note equation numbers (AN4165-D Eq. 39). Should our report do the same?                                              | (a) Formal: AN4165-D Eq. 39<br>(b) Informal: "per AN4165-D"<br>(c) None                                                | Option (a) is most professional and allows independent verification. Adds one footnote line per equation.                                                                 |
| 2 | <b>Starting knowledge level?</b><br>The reference documents assume familiarity with switching converters. Doc 2 defines duty cycle but does not derive it from first principles.                   | (a) Assume basic converter knowledge<br>(b) Assume PFC-specific knowledge<br>(c) Start from Ohm's Law                  | Recommendation: option (a). Matches the level of all four reference documents. Adds ~1 page of primer per major section.                                                  |
| 3 | <b>INSIGHT boxes — who writes them?</b><br>The most valuable INSIGHTs in Docs 1–3 come from engineering experience, not from the calculation engine.                                               | (a) Agent generates from calculation data<br>(b) Engineer provides; agent formats<br>(c) Agent draft; engineer reviews | Recommendation: option (c). Agent can draft many INSIGHTs from calculation data; engineer adds domain-experience insights.                                                |
| 4 | <b>Figures — dynamic or static?</b><br>All matplotlib plots are already dynamic. Block diagrams (ACM control architecture, circuit schematic) are not generated today.                             | (a) All dynamic matplotlib<br>(b) Static images for block diagrams only<br>(c) Add schematic via new code              | Recommendation: option (b). Embed static block-diagram PNGs from the specs/ folder for control architecture and schematic. Costs no development time.                     |
| 5 | <b>Corrections table — auto-generate or skip?</b><br>Doc 1 opens with a "corrections" table listing what was fixed from the previous design iteration. Our tool generates reports fresh each time. | (a) Auto-generate from design audit PASS/FAIL delta<br>(b) Fixed "this is revision 1.0" text<br>(c) Skip               | Recommendation: option (b) initially. Show "Initial release — no prior iteration" if no previous design stored. Future: auto-diff when a previous approved design exists. |

**Document Status:** Draft — For review before implementation begins. Based on complete study of all four reference documents, June 2026.

*Generated by PFC AI Design Agent documentation analysis session · June 2026 · Output:*

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